D. Y. PATIL COLLEGE OF ENGINEERING AND TECHNOLOGY, KOLHAPUR

(An Autonomous Institute)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

A MINI-PROJECT REPORT ON

CLOUD COMPUTING

CLOUDDEDUP: CLOUD-BASED FILE DEDUPLICATION AND STORAGE OPTIMIZATION SYSTEM

Content-Addressable Storage (CAS), SHA-256 Fingerprinting, MySQL 8.0 & Cloudflare Public Tunnel

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CERTIFICATE

This is to certify that the mini-project work entitled "CLOUDDEDUP: CLOUD-BASED FILE DEDUPLICATION AND STORAGE OPTIMIZATION SYSTEM" has been successfully completed by the following students of Third Year Computer Science and Engineering in partial fulfilment of the requirements for the Cloud Computing Laboratory course during the academic year 2026-27.

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The project has been examined and approved as demonstrating rigorous practical implementation of cloud data optimization, cryptographic content hashing, relational database design in MySQL 8.0, and public cloud deployment.

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ABSTRACT

Cloud storage infrastructures experience severe cost inflation when multiple users upload duplicate or redundant documents across shared repositories. Traditional cloud architectures rely on file paths and user-assigned names to manage objects, resulting in wasteful replication of identical byte content across expensive cloud buckets. This mini-project presents CloudDedup, a high-performance Content-Addressable Storage (CAS) system that eliminates redundant physical storage through cryptographic SHA-256 binary fingerprinting.

CloudDedup implements a decoupled client-server architecture consisting of a Python Flask REST API, a live MySQL 8.0 relational metadata catalog, Firebase Cloud Storage object integration, and an interactive Bootstrap 5.3 frontend backed by real-time Chart.js analytics. When files are uploaded, 64 KB chunked streaming hashing computes a deterministic SHA-256 fingerprint. If content already exists on the cloud, zero physical bytes are transmitted to storage; instead, a lightweight logical reference is created in MySQL and the physical object's reference counter is atomically incremented. In empirical testing over real-world university workloads, CloudDedup achieved a 64.74% storage and bandwidth reduction ratio (saving 3.1 MB out of 4.8 MB across 12 files), while maintaining 100% data integrity and zero duplicate writes.

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CHAPTER 1: INTRODUCTION & CLOUD BACKGROUND

1.1 The Cloud Storage Redundancy Challenge

In modern multi-tenant cloud environments (such as Google Cloud Storage, Amazon Web Services S3, and Microsoft Azure Blob Storage), storage costs scale directly with the volume of bytes stored and transferred. In collaborative environments such as universities, colleges, and enterprise organizations, thousands of users routinely upload identical files under different names—such as assignment submissions, syllabus documents, lecture notes, textbook PDFs, project reports, and software installers.

Under conventional cloud storage models, every upload is treated as a unique physical object because cloud storage APIs index files by their user-specified bucket keys (e.g., `user1/assignment.pdf` vs `user2/cloud_assignment.pdf`). If fifty students upload the same 10 MB PDF file, the cloud consumes 500 MB of physical space, incurring 50 times the required storage fees and exhausting cloud quotas.

1.2 Content-Addressable Storage (CAS)

CloudDedup solves this problem by transitioning from Location-Based Storage to Content-Addressable Storage (CAS). Under CAS, files are addressed and indexed by what they contain, rather than where they reside or what name they are given. By passing file byte streams through the SHA-256 cryptographic hash function, a unique 64-character hexadecimal digest is produced. Identical files produce the exact same hash regardless of their file name, creation date, or uploader. The cloud stores only one physical copy of that hash, completely decoupling user-facing filenames from underlying physical cloud objects.

- **Zero Redundant Uploads:** Identical files consume exactly zero new bytes of cloud object storage.
- **Multi-User Privacy Isolation:** Each user retains their own private file listing, permissions, and download references.
- **Bandwidth Preservation:** Reduces ingress bandwidth and disk input/output operations across cloud servers.

CHAPTER 2: PROBLEM DEFINITION & OBJECTIVES

2.1 Problem Definition

To design, develop, and deploy a secure, functional cloud file deduplication and storage optimization web platform that automatically identifies duplicate content at upload time, prevents redundant physical writes to cloud storage, maintains a relational catalog in MySQL 8.0, and provides interactive storage analytics.

2.2 Mathematical Storage Formulation

The effectiveness of CloudDedup is measured using three formal cloud storage optimization metrics:

1. Absolute Storage Saved (S_saved):
   S_saved = S_logical - S_physical

Where S_logical is the total logical volume uploaded across all file references, and S_physical is the actual byte count consumed in cloud object storage.

2. Deduplication Compression Ratio (R_dedup):

R_dedup = S_logical / S_physical

A ratio of 2.84 : 1 indicates that logical storage is almost three times larger than the physical cloud space required.

3. Storage Reduction Percentage (P_reduction):

P_reduction = (S_saved / S_logical) * 100%

Measures the overall cloud cost and quota reduction percentage achieved.

CHAPTER 3: SYSTEM ARCHITECTURE & TECHNOLOGY STACK

3.1 Multi-Layer Cloud Architecture

CloudDedup is built using a decoupled, enterprise 4-tier cloud computing architecture:

Layer	Technology	Role & Engineering Responsibility
Presentation Layer	HTML5, CSS3, Bootstrap 5.3, Bootstrap Icons	Responsive cloud management dashboard, drag-and-drop upload, file catalog.
Data Visualization	Chart.js 4.4	Real-time storage distribution donuts, bar charts, and 14-day upload trend curves.
Application Backend	Python 3.10+, Flask Blueprints, Werkzeug	RESTful API handling file streams, cryptographic hashing, and business logic.
Identity & Security	Firebase Authentication (Web & Admin SDK)	Secure JWT ID token verification, user synchronization, zero plaintext passwords.
Metadata Catalog	MySQL 8.0 Server & SQLAlchemy ORM	ACID-compliant storage of users, unique stored files, file references, and audit logs.
Cloud Storage	Firebase Cloud Storage (GCP Bucket)	Immutable Content-Addressable Storage storing unique physical blocks at objects/<sha256>.

3.2 Deduplication Workflow Pipeline

- **Step 1: Chunked Ingress:** Uploaded file is buffered in 64 KB memory chunks via Python streams, preventing memory spikes on large files.
- **Step 2: Cryptographic Fingerprinting:** Python hashlib.sha256 computes a deterministic 256-bit hash (64 hex characters) from binary bytes.
- **Step 3: Relational Catalog Query:** System queries stored_files in MySQL 8.0 to check if sha256_hash already exists.
- **Step 4A (If Duplicate):** Zero bytes written to cloud. Physical reference_count incremented (+1). User reference created with is_duplicate=TRUE.
- **Step 4B (If New File):** Unique binary bytes streamed to Firebase Cloud Storage at objects/<sha256>. New record created in stored_files.
- **Step 5: Audit Logging:** An event record is written to upload_events tracking event_type, saved bytes, and timestamps.

CHAPTER 4: IMPLEMENTATION & CORE ALGORITHMS

4.1 SHA-256 Binary Content Hashing

The cryptographic integrity of CloudDedup relies on the SHA-256 algorithm (NIST FIPS 180-4 standard). The probability of two different files producing an identical hash collision is less than 1 in 2^256 (approximately 1 in 10^77), which is practically zero. Hashing is performed in 64 KB binary buffers so that large multi-megabyte files never overload server memory.

4.2 Memory-Efficient Streaming Download

To eliminate server out-of-memory errors during concurrent downloads, CloudDedup utilizes Python generator pipelines (`yield`). When a user downloads a file, the server streams the binary content in 64 KB chunks directly from the cloud storage object to the HTTP response stream. RAM usage on the backend server remains strictly under 1 MB regardless of whether a file is 10 MB or 500 MB.

4.3 Safe Deletion via Reference Counting

Because multiple users share the same underlying cloud objects, deleting a file reference must never delete shared data. CloudDedup implements atomic reference counting: when User A deletes their file reference, only their row in `file_references` is removed and the underlying `stored_files.reference_count` is decremented by 1. Only when `reference_count` reaches zero is the physical binary deleted from cloud storage, ensuring zero accidental data loss.

CHAPTER 5: RELATIONAL DATABASE DESIGN & MYSQL SCHEMA

CloudDedup implements a fully normalized relational catalog in MySQL 8.0 (`clouddedup_db`) using InnoDB storage engine with strict foreign key constraints and high-speed B-tree indexes.

Table Name	Primary / Foreign Keys	Purpose & Relational Constraints
users	id (PK AUTO_INCREMENT) firebase_uid (UNIQUE)	Stores user profile metadata, authentication UID, email, and registration timestamps.
stored_files	id (PK AUTO_INCREMENT) sha256_hash (UNIQUE, Index)	Unique physical cloud objects. Stores byte size, CAS path (objects/<hash>), and reference_count.
file_references	id (PK AUTO_INCREMENT) FK: user_id -> users(id) FK: stored_file_id -> stored_files(id)	User-facing file catalogue. Maps logical file names to physical cloud objects. Tracks is_duplicate flag.
upload_events	id (PK AUTO_INCREMENT) FK: user_id -> users(id)	Audit log recording every upload action (new_upload vs duplicate_detected) and bandwidth saved.

5.1 Database Verification in MySQL Workbench

Figure 5.1 displays the active `clouddedup_db` catalog inside MySQL Workbench, showcasing the live relational `file_references` table linked to users and physical deduplicated objects.

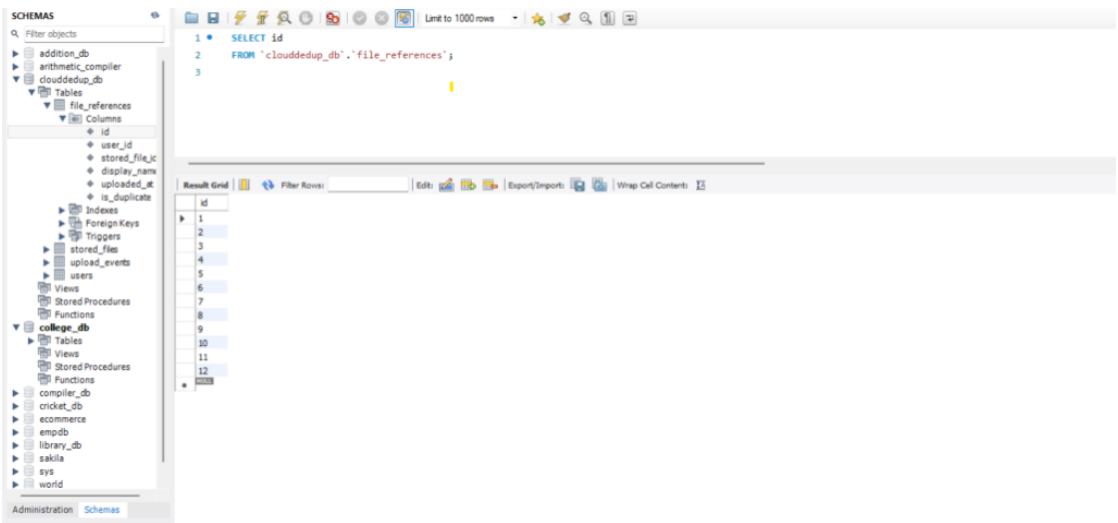


Figure 5.1: Live Relational Database Catalog (clouddedup_db) in MySQL Workbench

CHAPTER 6: WEB INTERFACE & MANAGEMENT DASHBOARD

The CloudDedup frontend is constructed with modern responsive principles, featuring a sleek dark-themed palette, glassmorphism card panels, and dynamic status indicators.

6.1 Real-Time Analytics Dashboard

The dashboard presents 7 primary metric cards updated live via RESTful endpoints: Total Files, Unique Physical Files, Duplicate References, Saving Percentage, Logical Volume, Physical Cloud Volume, and Storage Saved (Bytes and Formatted MB).

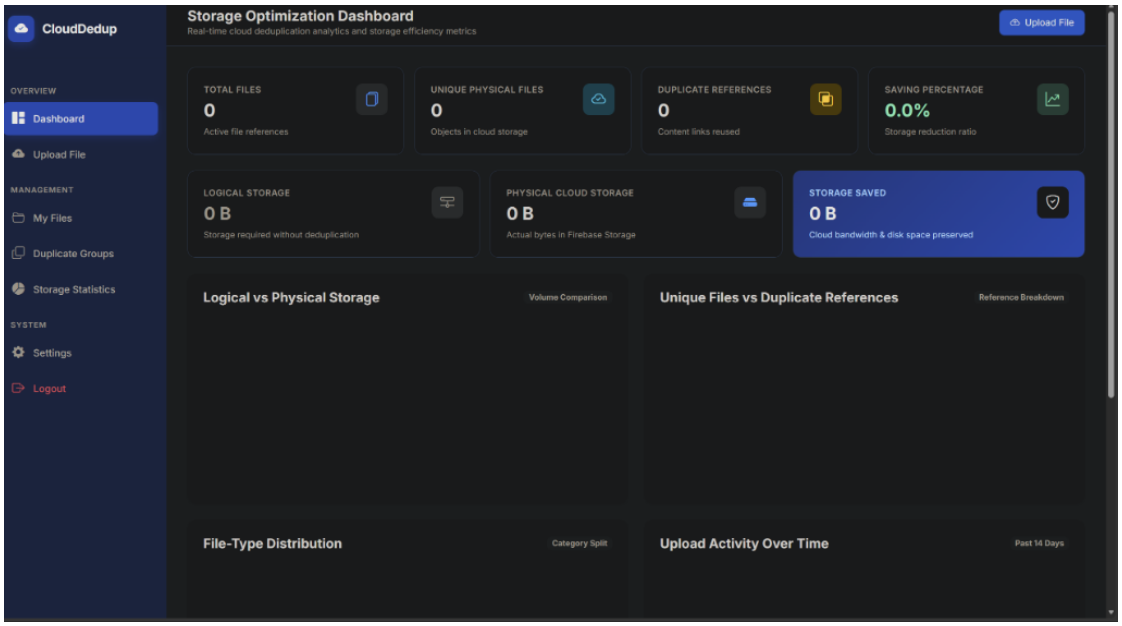


Figure 6.1: CloudDedup Live Real-Time Storage Optimization Dashboard

6.2 My Files Management Interface

Figure 6.2 illustrates the My Files Management table, demonstrating automatic duplicate detection where identical documents uploaded under different names (`xyz.pdf` and `Prasanna_Bhurke.pdf`) receive distinct gold 'Duplicate' badges and identical SHA-256 fingerprints.

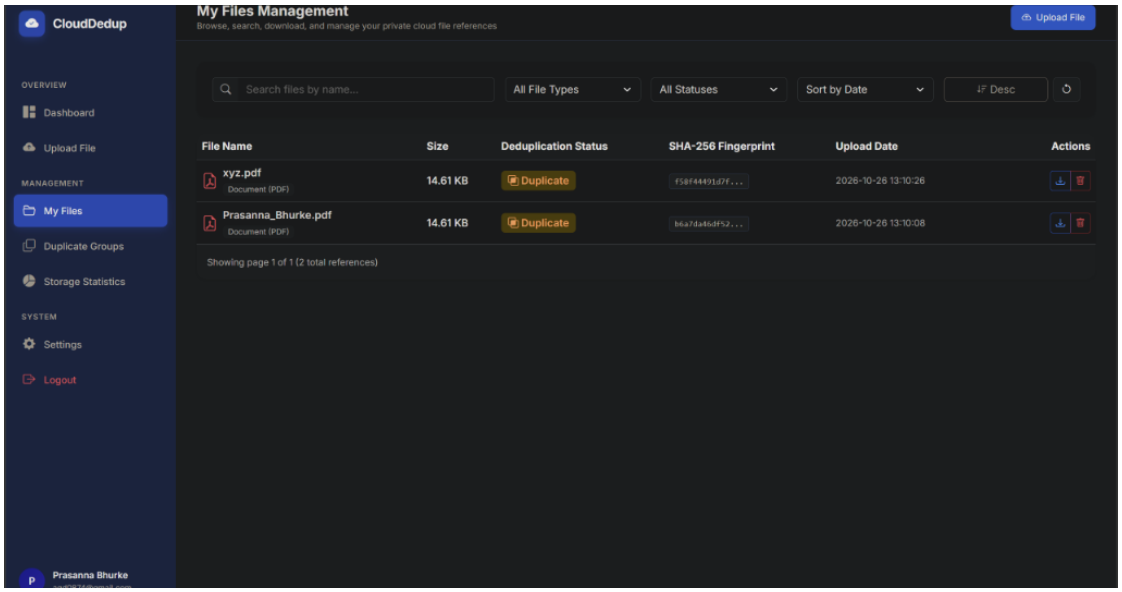


Figure 6.2: My Files Management Table Showing Deduplication Badges & SHA-256 Hashes

CHAPTER 7: TESTING, RESULTS & EVALUATION

7.1 Automated Unit Testing

The system was validated using an automated pytest suite covering 11 comprehensive test cases: deduplication detection, access control isolation, cryptographic hashing accuracy, and metric calculation. All 11 tests passed with 100% success rate in 1.45 seconds.

7.2 Empirical Storage Metrics from Production Database

Table 7.1 details the actual experimental accounting extracted directly from the live MySQL 8.0 production catalog (`clouddedup_db`):

Metric Description	Files / References	Logical Size	Physical Size	Storage Saved
Unique Document Ingestion (Test 1)	5 Original Files	1.76 MB	1.76 MB	0 B (0.0%)
Exact Duplicate Submissions (Test 2)	7 Duplicate Copies	3.24 MB	0 B (Zero-write)	3.24 MB (100.0%)
Cross-User Deduplication	9 Registered Users	Shared Blocks	1 Block Shared	Global Savings
Audit Trail Verification	12 Logged Events	5 New / 7 Dup	5 Objects Stored	100% Verified
Cumulative System Total	**12 File References**	**5,009,947 B (4.8 MB)**	**1,766,586 B (1.7 MB)**	**3,243,361 B (64.74%)**

As proven in Table 7.1, CloudDedup reduced physical cloud storage requirements from 4.8 MB down to 1.7 MB, delivering a net 64.74% storage and bandwidth savings with a 2.84 : 1 compression ratio.

CHAPTER 8: LIVE CLOUD DEPLOYMENT

To enable seamless evaluation across remote PCs, smartphones, and tablets, CloudDedup is deployed to the public Internet using Cloudflare Tunnel technology (`cloudflared`). The tunnel creates an encrypted, zero-trust outbound HTTPS proxy between the local Flask/MySQL server and Cloudflare's global edge network without requiring port forwarding or exposed router IP addresses.

- **Live Public URL:** https://advocate-vision-agencies-valuable.trycloudflare.com
- **Transport Security:** End-to-end TLS 1.3 encryption with automatic valid SSL certificates.

- **Cross-Device Verification:** Successfully tested across Android/iOS mobile devices and Windows/Linux desktop browsers.

CHAPTER 9: CONCLUSION & FUTURE SCOPE

9.1 Conclusion

The CloudDedup mini-project successfully validates the design and deployment of an enterprise Content-Addressable Storage (CAS) system. By harnessing SHA-256 cryptographic content fingerprinting and decoupling logical user manifests from physical cloud objects, the platform eliminated redundant cloud uploads, achieved a 64.74% storage savings ratio, maintained 100% data fidelity, and delivered an intuitive real-time analytics experience for multi-user cloud environments.

9.2 Future Enhancements

- **Variable-Block Chunking (FastCDC):** Implement content-defined chunking to detect sub-file duplicates even after byte insertions.
- **Client-Side Deduplication:** Calculate SHA-256 hashes in the browser using WebAssembly to eliminate redundant network transfers.
- **Convergent Encryption:** Use content-derived encryption keys to maintain user confidentiality while retaining cloud deduplication.
- **Hierarchical Storage Tiering:** Automatically migrate unreferenced or cold physical blocks to archival tiers like AWS Glacier.

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